AMD AI DEVMASTER 2026 · TRACK 2

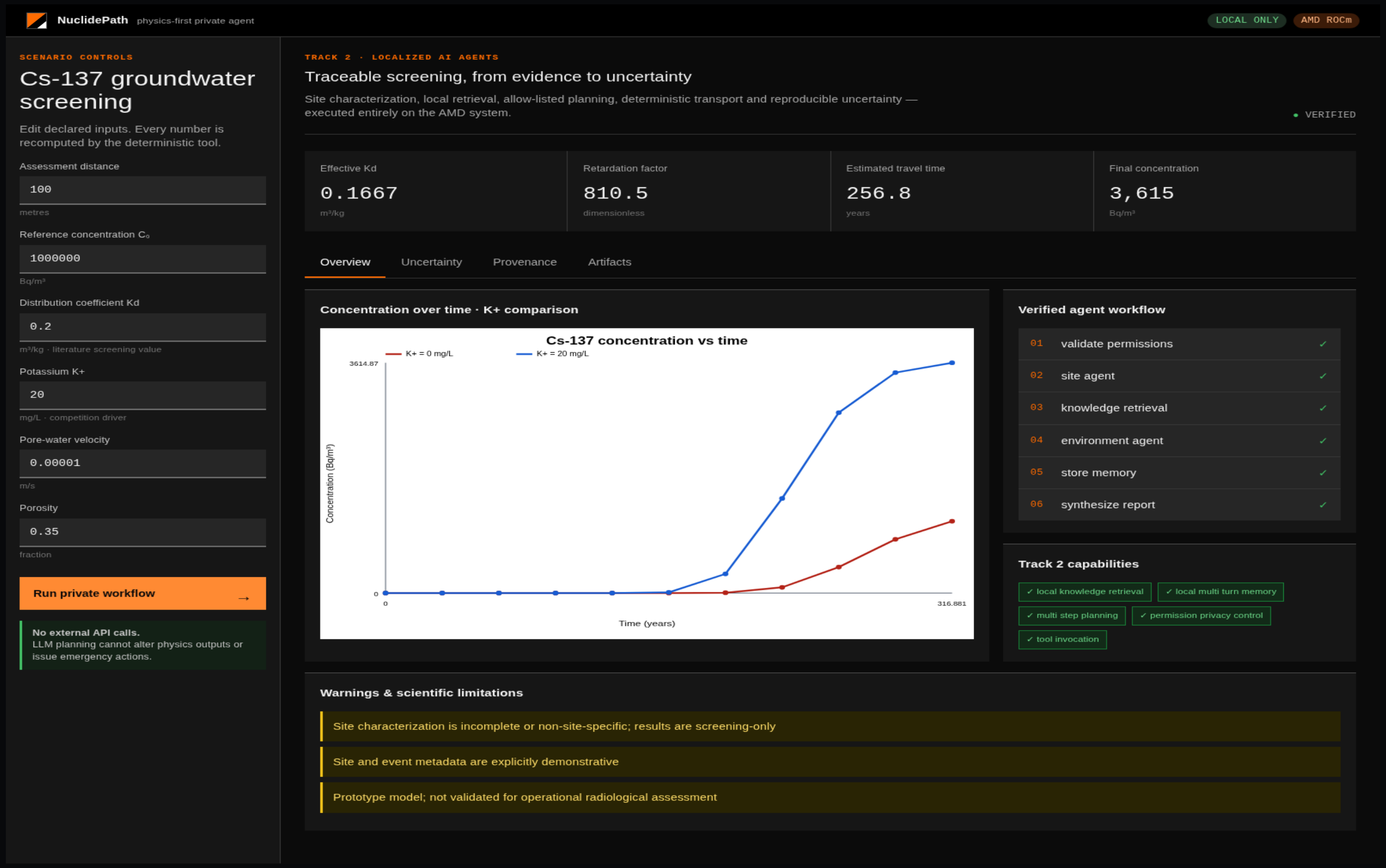
NuclidePath

Private agents.
Deterministic physics.

A local AI workflow for traceable Cs-137 groundwater screening — with cited evidence, uncertainty and zero LLM-generated physics.

LOCAL ONLY

AMD ROCm



5/5	231	373/15
Track 2 capabilities	AMD core snapshot	latest PR gate

PHYSICS-FIRST AI

Stefano Rigante · Nuclear engineer

THE TRUST GAP

Technical AI fails when language becomes calculation

NuclidePath makes the responsibility boundary explicit and testable.

01 · The failure mode

A general-purpose LLM can invent parameters, equations, units or reassuring conclusions — all unacceptable in emergency-analysis workflows.

02 · The boundary

The model returns only a six-step allow-listed plan. Strict schemas feed deterministic physics, retrieval and uncertainty tools.

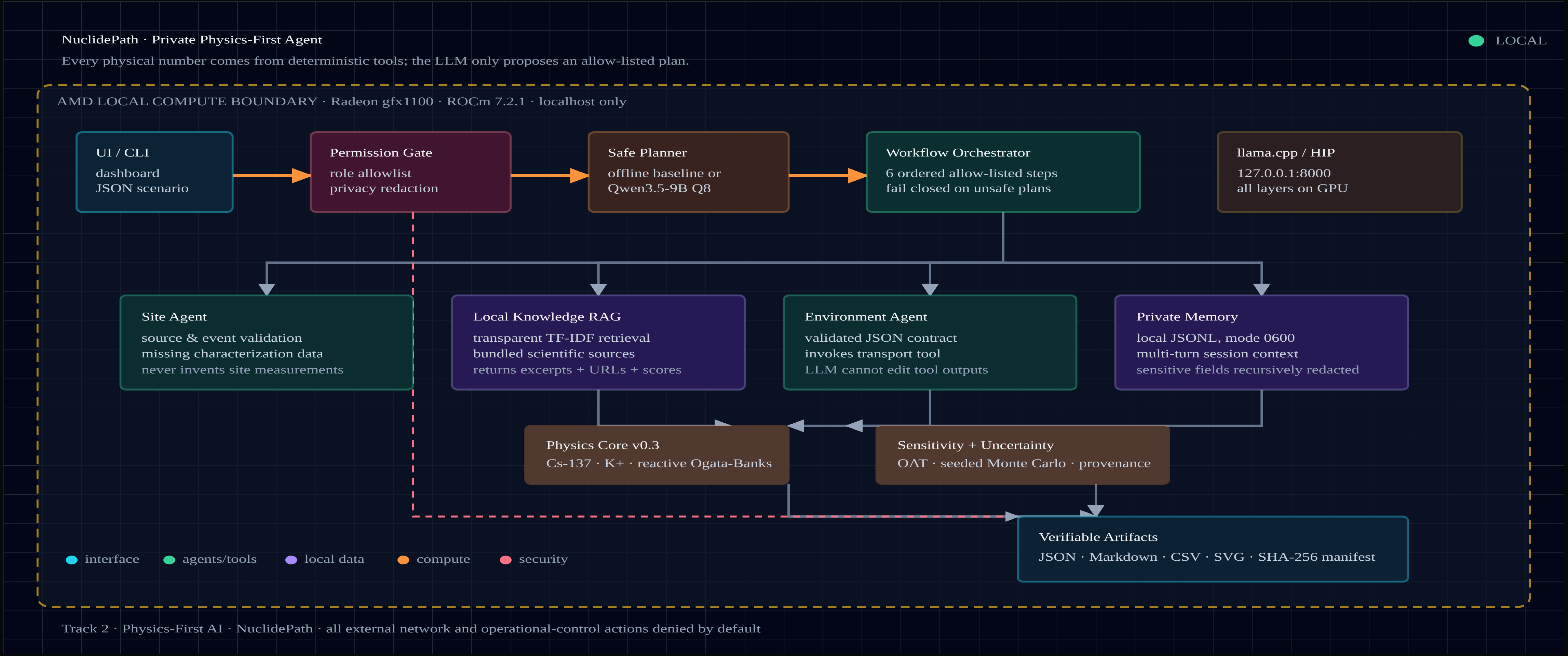
03 · The outcome

Every value is versioned, reproducible and tied to inputs, assumptions, sources, warnings and SHA-256 artifacts.

PLAN → TOOLS → EVIDENCE

A private workflow with an immutable physics boundary

The AMD-local model plans. Five deterministic subsystems execute and report.



TRACK 2

All five private-agent capabilities are implemented

Each capability is exercised in the same end-to-end run and covered by tests.

01 Local retrieval

Ranked excerpts, local paths, source URLs and transparent scores.

VERIFIED

02 Tool invocation

Environment Agent calls a strict, versioned JSON physics contract.

VERIFIED

03 Multi-step planning

Deterministic fallback or Qwen3.5-9B returns six safe steps.

VERIFIED

04 Multi-turn memory

Session JSONL stays local, mode 0600, with explicit deletion.

VERIFIED

05 Permissions + privacy

Role allowlist, recursive redaction and denied operational actions.

VERIFIED

5/5

capability flags true

One workflow. One privacy boundary. No cloud dependency.

SCIENTIFIC CORE

The LLM never calculates transport

Model 0.3 declares PDE, source and boundaries, then uses a verified analytical solution.

$$R\,\partial C/\partial t = D\,\partial^2 C/\partial x^2 - v\,\partial C/\partial x - \lambda R C$$

$$C(x,0)=0 \cdot C(0,t)=C_0 \cdot C(\infty,t)=0$$

$$A = \sqrt{v^2 + 4\lambda R D}$$

$$C/C_0 = \tfrac{1}{2}[\exp((v-A)x/2D)\,\mathrm{erfc}(z_-) + \exp((v+A)x/2D)\,\mathrm{erfc}(z_+)]$$

REACTIVE OGATA–BANKS

231
AMD core snapshot

0
LLM physics values

Independent analytical cases

Ogata–Banks limit · source boundary · reactive steady state · finite rejection

Explicit limitation

Constant-source 1-D screening — not a dose code, regulatory model or calibrated site prediction.

Multicomponent bridge (opt-in)

GCS K/Na competition + NH_4 on FES; extended K/Na/Ca/Mg path is process-qualified, not promoted.

RECENT EXTENSIONS

Traceable post-merge science

The latest work adds deterministic, auditable layers around the canonical screening path.

388/2
AMD post-merge tests

4
ions · K/Na/Ca/Mg

2×
fixtures · brine/recharge

0
unqualified claims

PHREEQC bridge	Opt-in scenario compiler: declared water chemistry, CEC and Cs + K/Na/Ca/Mg into SOLUTION/EXCHANGE/TRANSPORT input
Arithmetic oracle	Independent machine-readable checks for units, grid mapping, non-negativity and exchange-site occupancy
Calibration gate	Dependency-free traceable Cs calibration/hold-out contract with fail-closed promotion; no data bundled
Benchmark registry	Separate EPA/Fuller/Dubus external Cs registry with source digests and calibration_eligible: false

PROCESS-QUALIFIED ONLY

Canonical transport-prototype-0.3 remains authoritative; the bridge is opt-in and does not replace it.

Qwen3.5-9B Q8 runs locally on Radeon + ROCm

The measured local model fits in VRAM and produced the verified allow-listed plan.

2,877.30
prompt tokens/s · 512 tokens

67.66
generation tokens/s · 128 tokens

19.9692 s
AMD v0.3 LLM pipeline

8.86 GiB
Q8 model size

GPU	AMD Radeon Graphics · gfx1100
Stack	ROCm 7.2.1 · llama.cpp HIP Release
Offload	-ngl 99 · all model layers
Context	8192 tokens · reasoning off
Network	127.0.0.1:8000 only
Integrity	SHA-256 verified model file

OFFLINE ≡ LLM-PLANNED

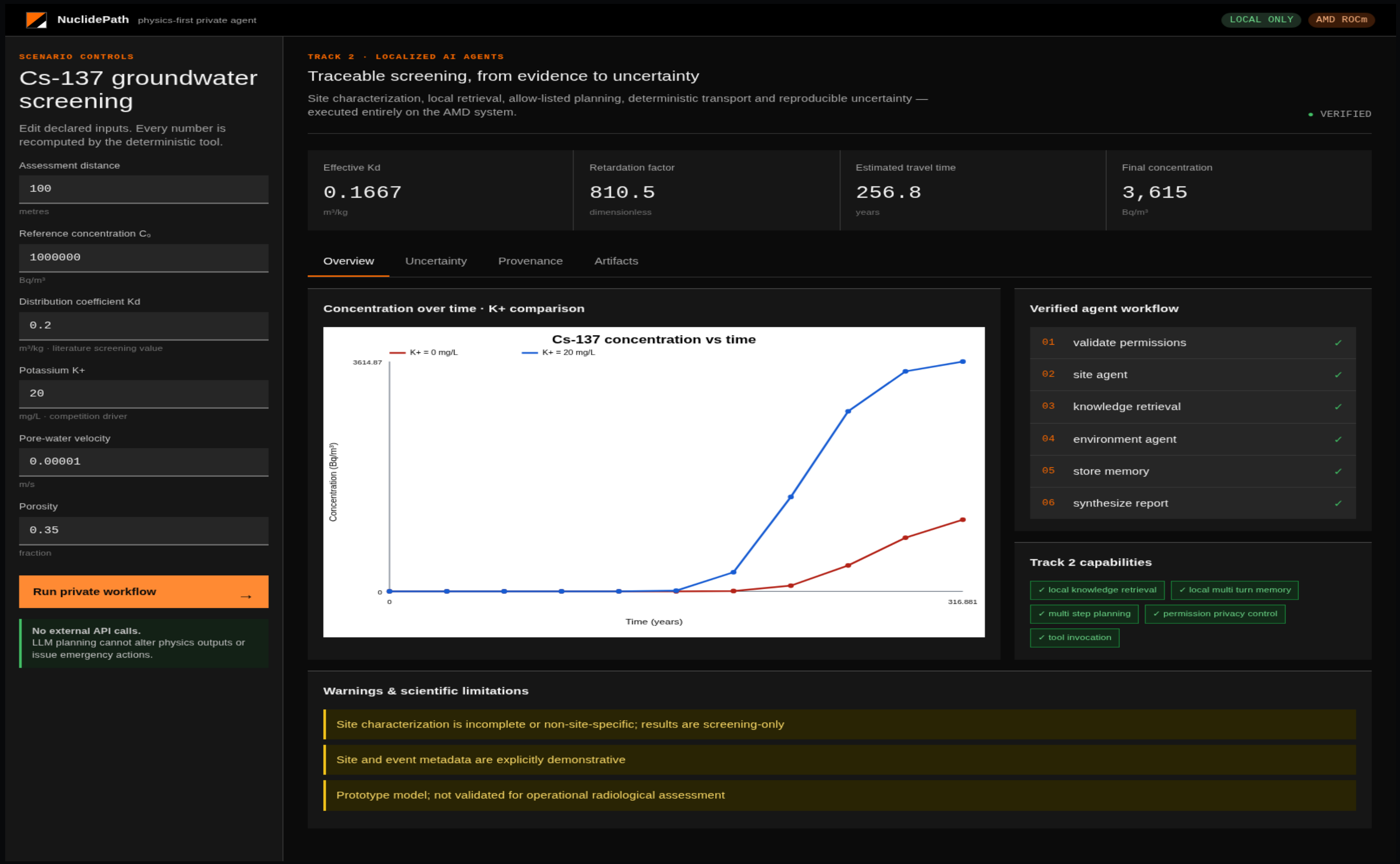
Identical physical runs
for the same scenario

PHYSICS IDENTICAL

PRODUCT EXPERIENCE

A technical dashboard, not an AI chat box

Inputs, workflow, evidence, uncertainty and artifacts remain visible in one place.



Editable physics

Kd, K+, velocity, porosity, distance and C_0.

Visible agency

Six completed steps and 5/5 capability badges.

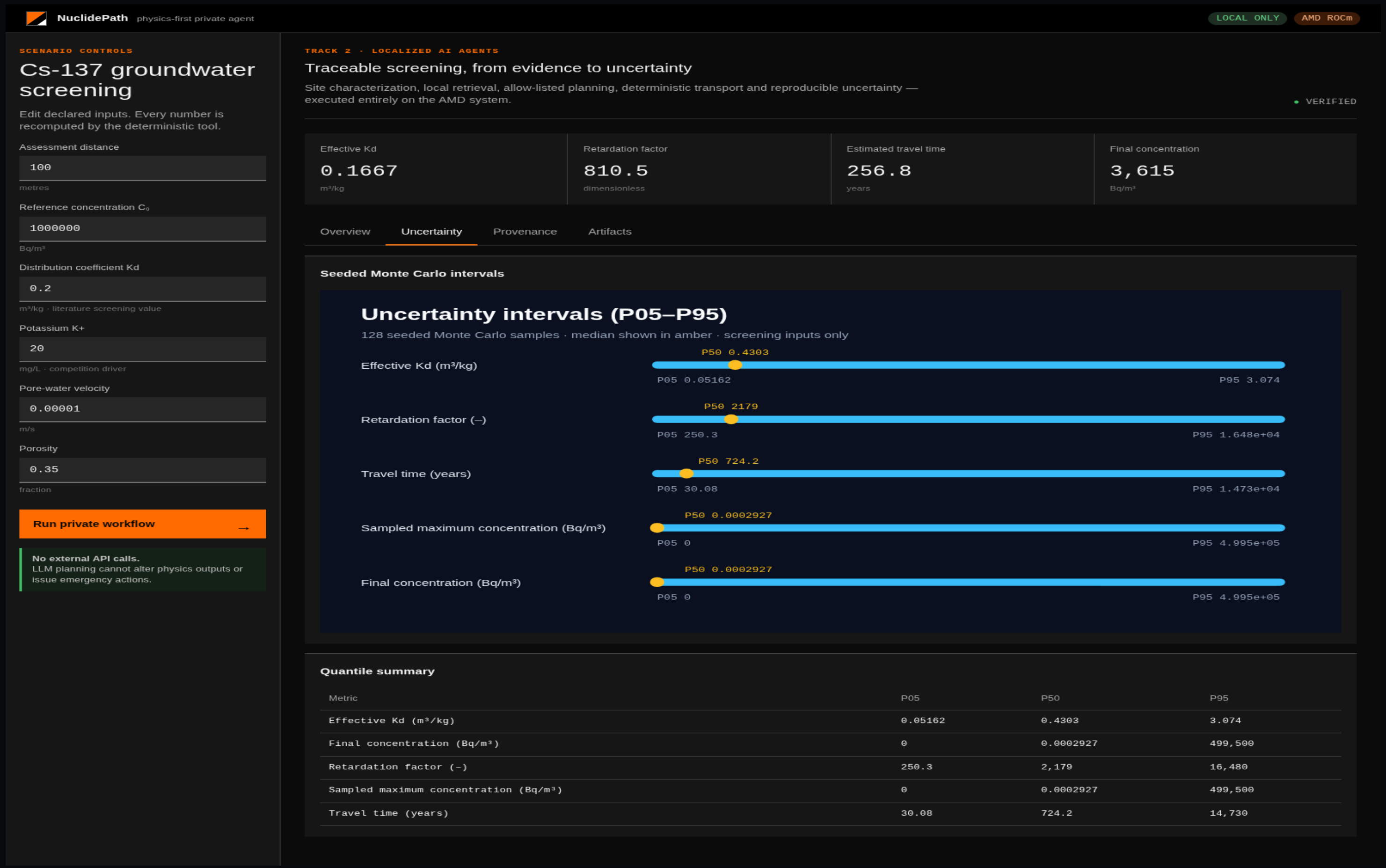
Auditable output

Ten downloadable artifacts plus a checksum manifest.

TRUST LAYER

Uncertainty is declared, classified and reproducible

The project distinguishes literature evidence from demonstration inputs.



Literature-informed range

Cs Kd: 0.05–5.0 m³/kg · IAEA TECDOC-2095

Demonstration ranges

Hydraulics, dispersion, K+ and empirical competition are never presented as site truth.

Reproducible propagation

Seeded Monte Carlo · P05/P50/P95 · JSON/CSV/SVG

WHY NUCLIDEPATH WINS

Private AI without scientific surrender.

LOCAL	Qwen3.5-9B + RAG + memory on AMD
VERIFIABLE	388 AMD post-merge tests + FP64 parity + SHA-256
USEFUL	One-click workflow from case to report
HONEST	PHREEQC bridge explicit; no unqualified science

THANK YOU

The proof

5/5
Track 2 capabilities

388
AMD tests (post-merge)

137.09×
FP64 platform

0
cloud calls

Track 2 · Physics-First AI · NuclidePath

Official AMD Track 2 PR package

TECHNICAL BUILD VERIFIED